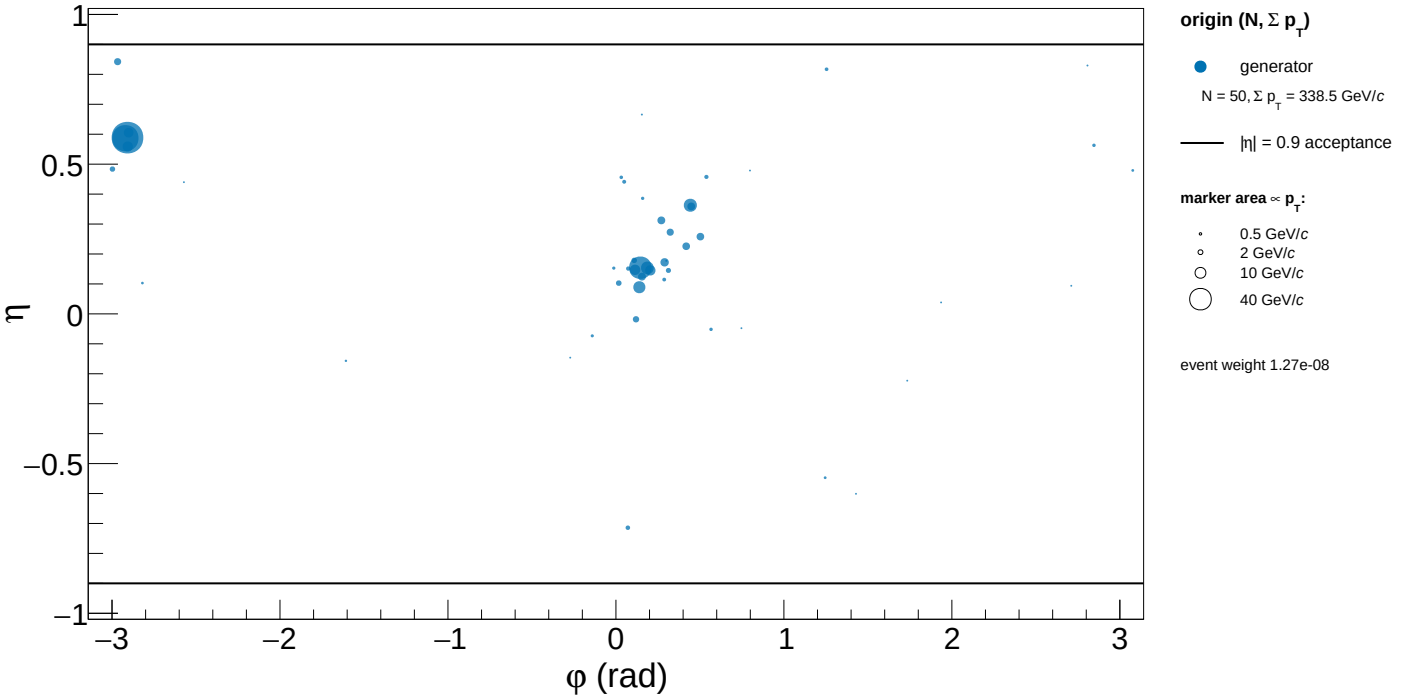
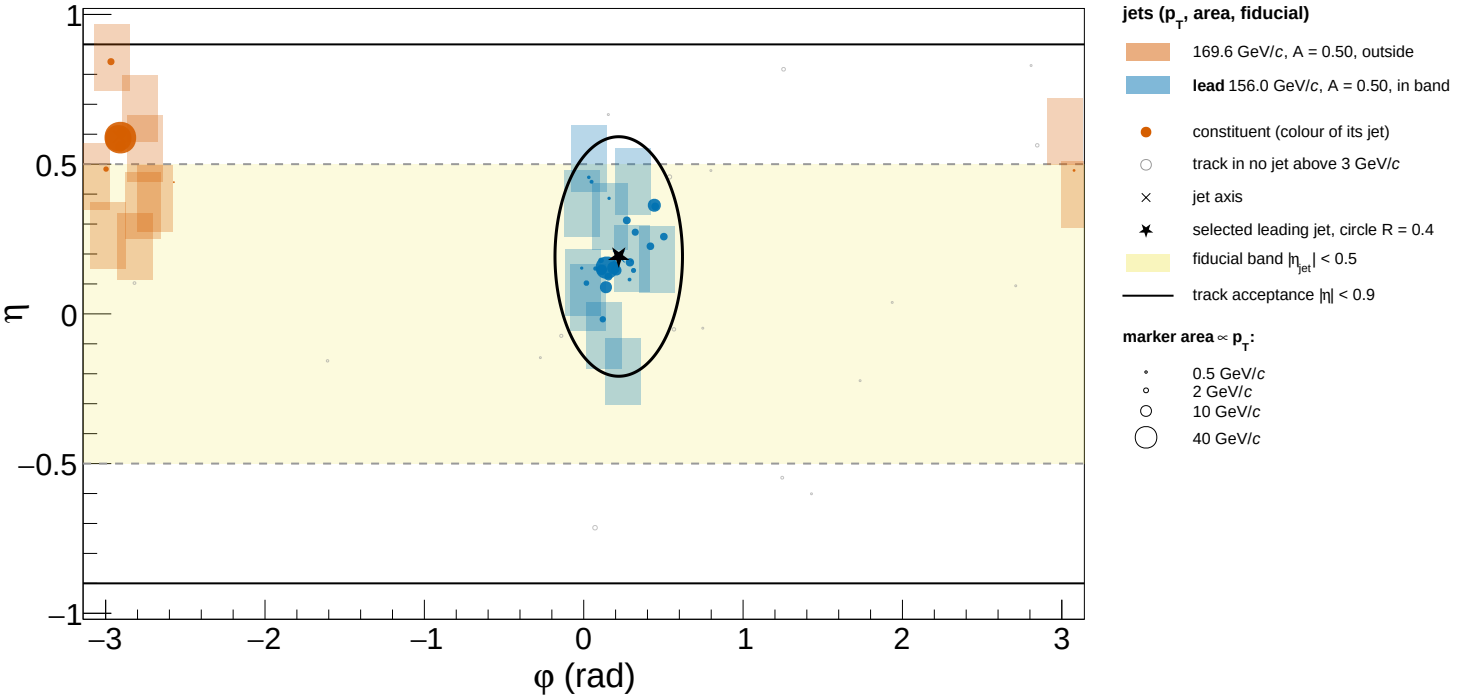
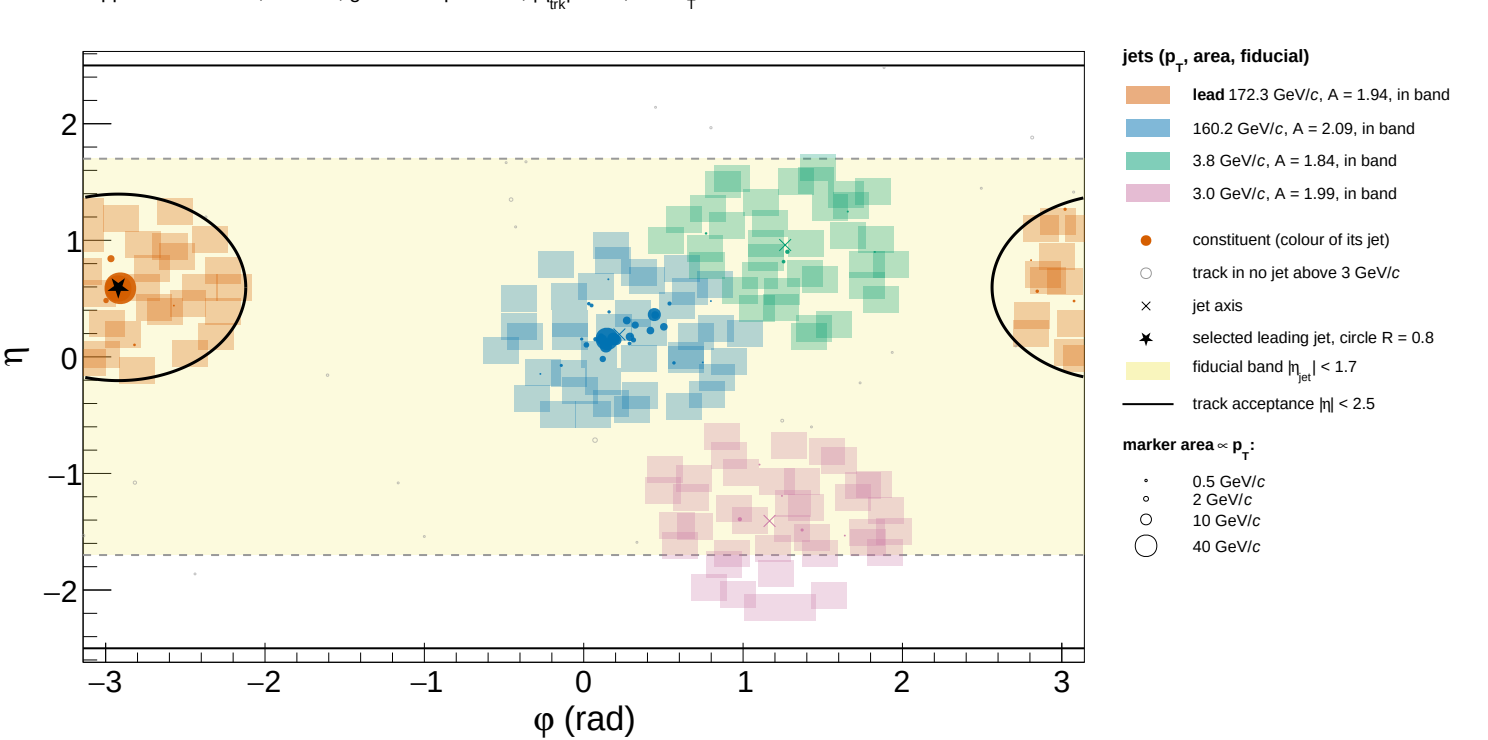


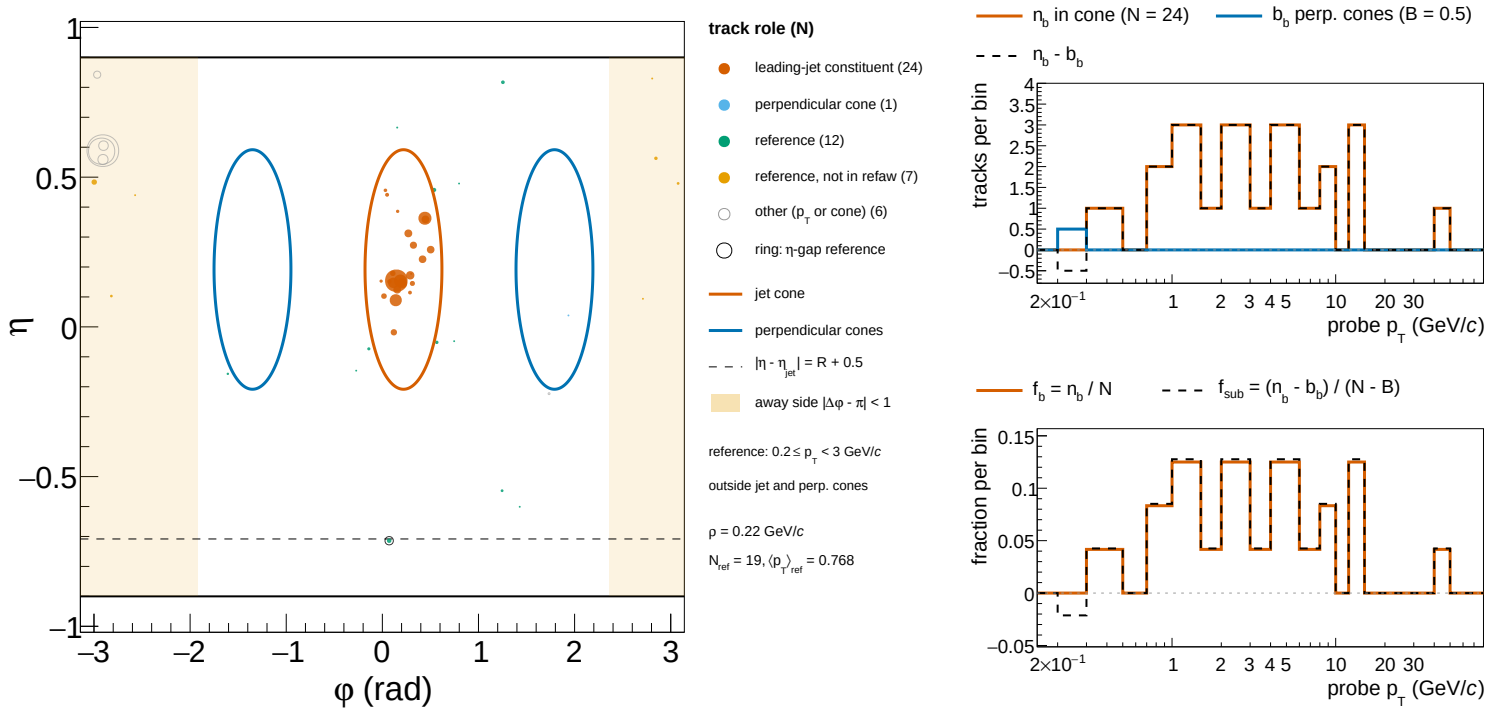
(a) Charged tracks entering the jet clustering ($p_T > 0.15$ GeV/c); marker area $\propto p_T$, colour = origin of the particle.



(b) Anti-k$_T$ R = 0.4 jets with p_T > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 0.5$, shaded) circled.

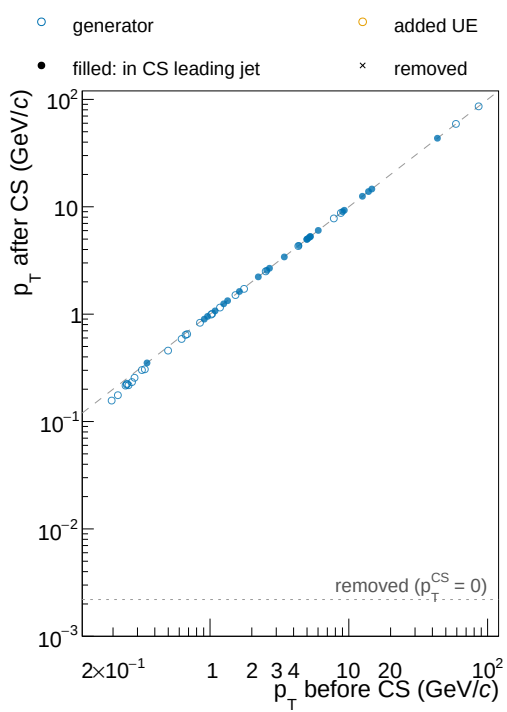


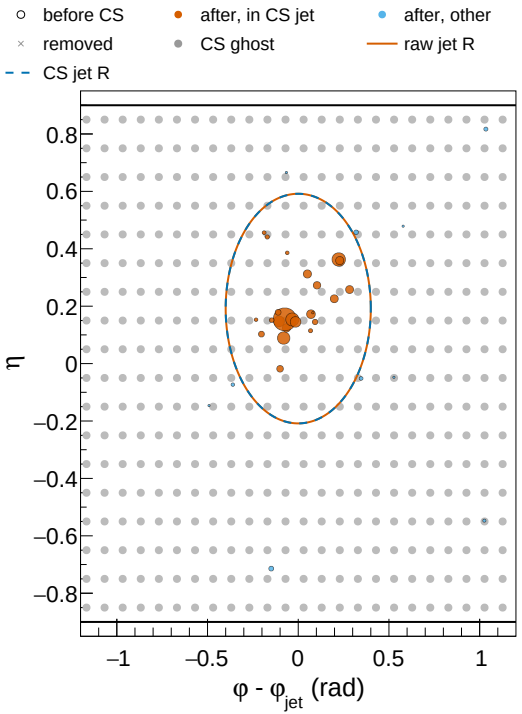
(b) Anti-k$_T$ R = 0.8 jets with p$_T$ > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 1.7$, shaded) circled.



(c) Jet cone and perpendicular cones ($\varphi_{jet} \pm \pi/2$), track roles; right: in-cone counts n_b, perpendicular-cone estimate b_b, $n_b - b_b$ and the fractions f_b, f_{sub} per probe-p_T bin.

PYTHIA 8 pp $\sqrt{s}$ = 5.36 TeV, event 4, generator particles, $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4





leading jet, R = 0.4

raw p_T = 156.02 GeV/c

area A = 0.505, ρ = 0.22 GeV/c

ρA-corrected p_T = 155.92 GeV/c

CS jet p_T = 155.91 GeV/c

truth (no added UE) p_T = 156.02 GeV/c

tracks in the acceptance

Σ p_T before CS = 338.5 GeV/c

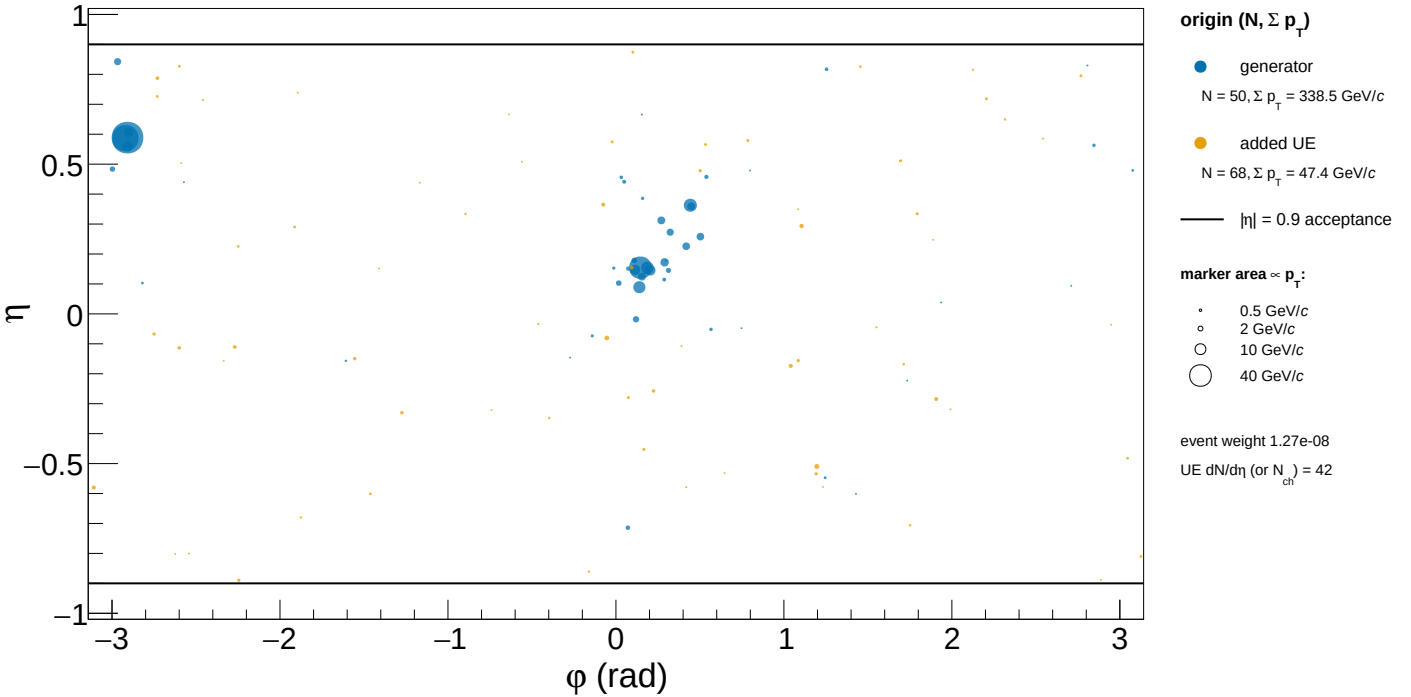
Σ p_T after CS = 337.7 GeV/c

removed: 0 of 50 tracks (0.0 GeV/c)

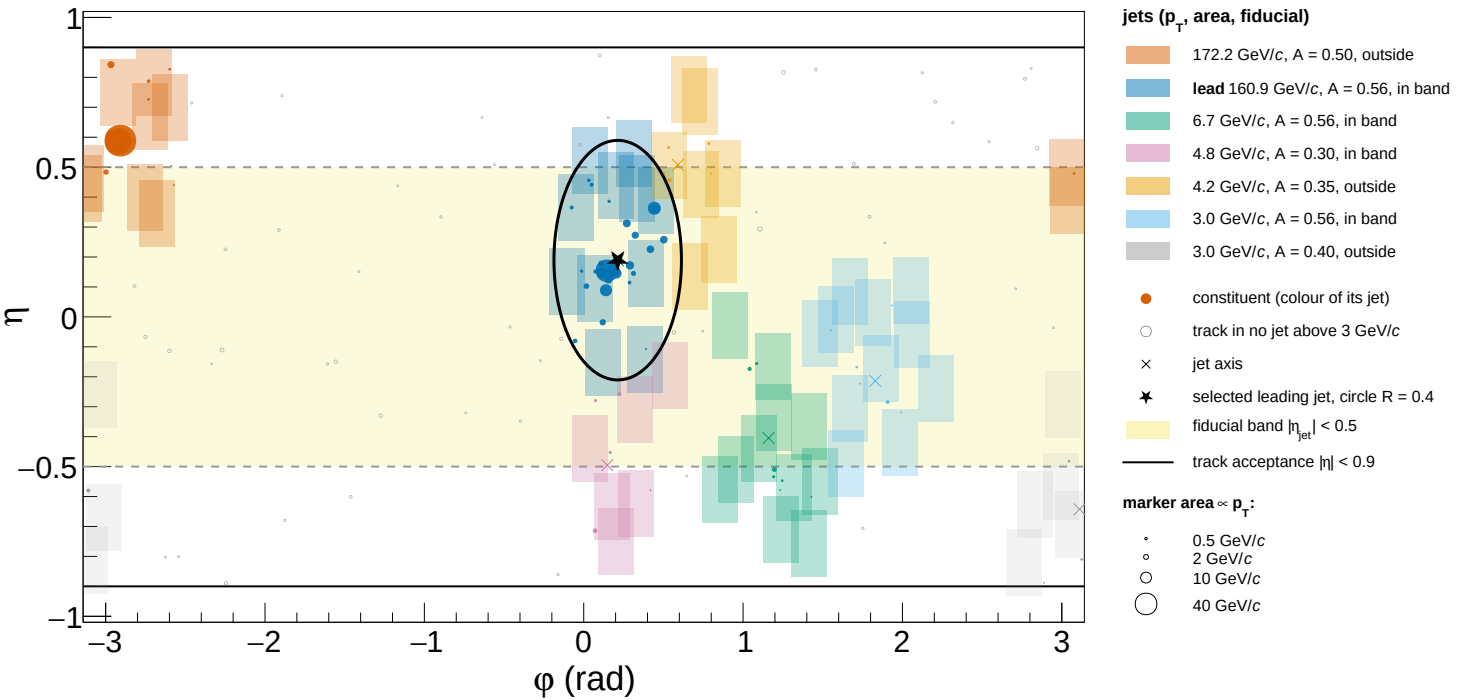
subtracted Σ p_T = 0.8 GeV/c

($ρ_{CS}$ × acceptance = 2.4 GeV/c;
ghosts farther than $ΔR_{max}$ keep their p_T)

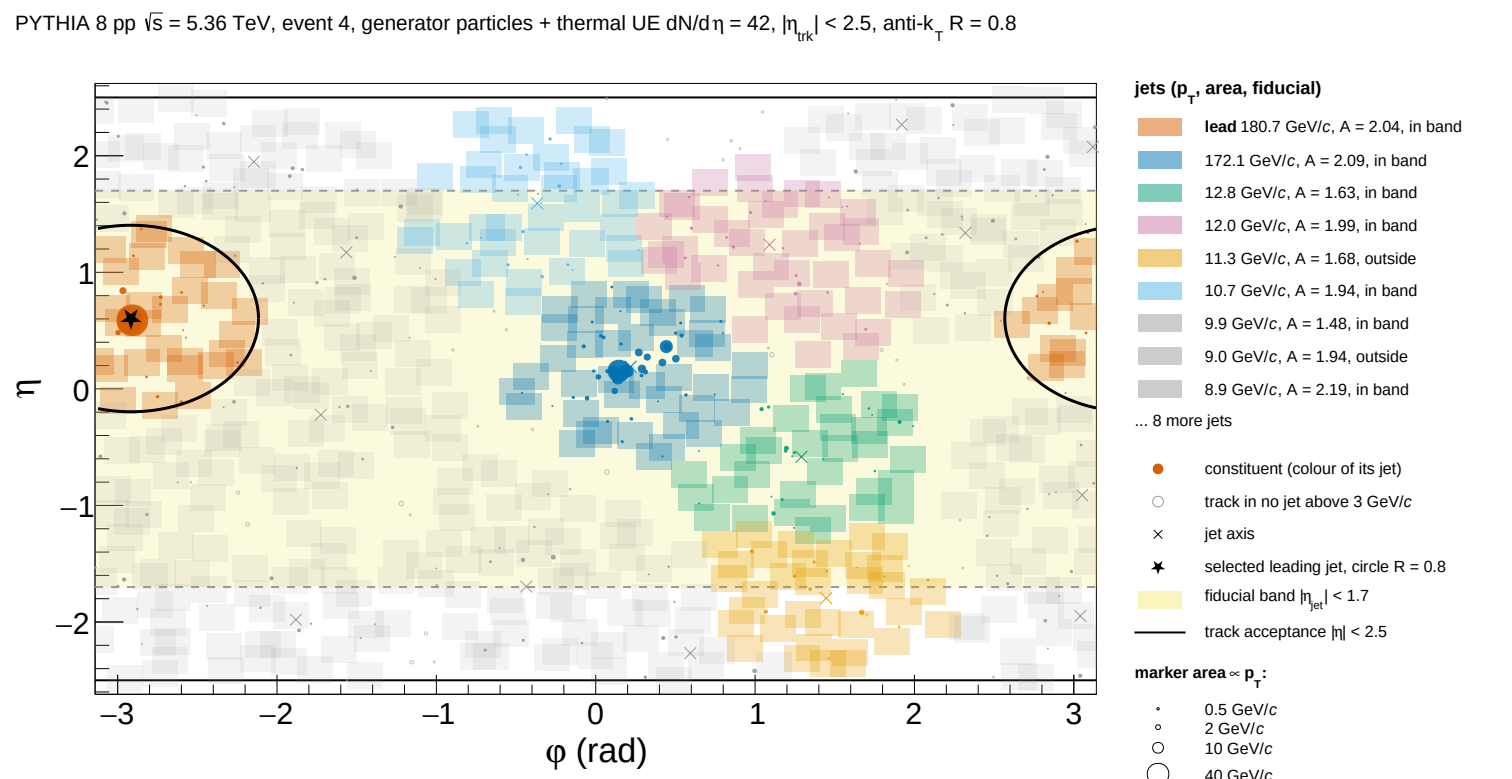
(d) Event-wise constituent subtraction ($ΔR_{max}$ = 0.25, α = 0, ghost area 0.01, $ρ_{CS}$ = 0.22 GeV/c): track p_T before/after, CS ghost grid near the jet, jet p_T definitions.



(a) Charged tracks entering the jet clustering (p_T > 0.15 GeV/c); marker area ∝ p_T, colour = origin of the particle.

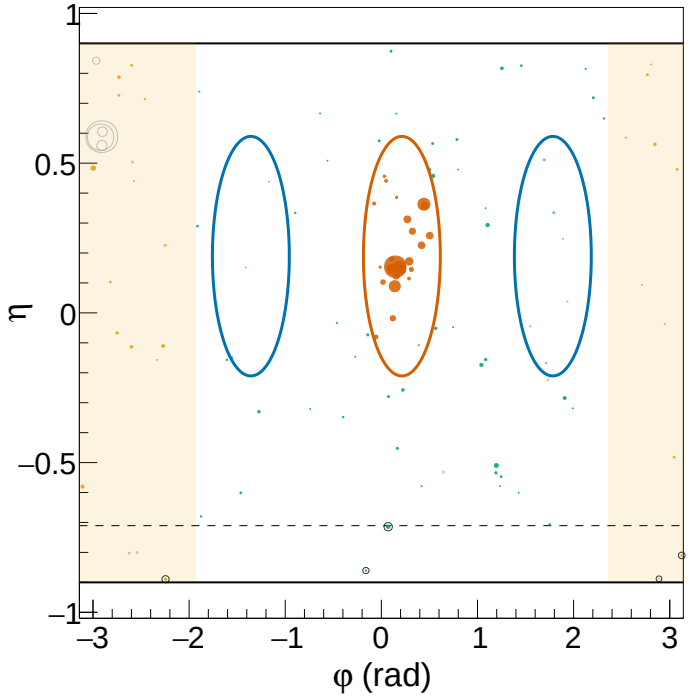


(b) Anti-k$_T$ R = 0.4 jets with p$_T$ > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 0.5$, shaded) circled.



(b) Anti-k$_T$ R = 0.8 jets with p$_T$ > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 1.7$, shaded) circled.

PYTHIA 8 pp $\sqrt{s}$ = 5.36 TeV, event 4, generator particles + thermal UE dN/dη = 42, $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



track role (N)

- leading-jet constituent (28)
- perpendicular cone (9)
- reference (47)
- reference, not in refaw (24)
- other (p_T or cone) (10)
- ring: η-gap reference

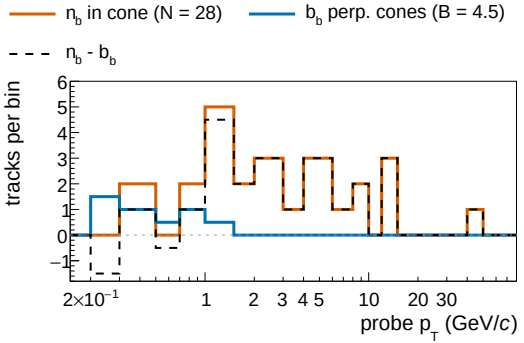
- jet cone
- perpendicular cones
- $|\eta - \eta_{jet}| = R + 0.5$
- away side $|\Delta\varphi - \pi| < 1$

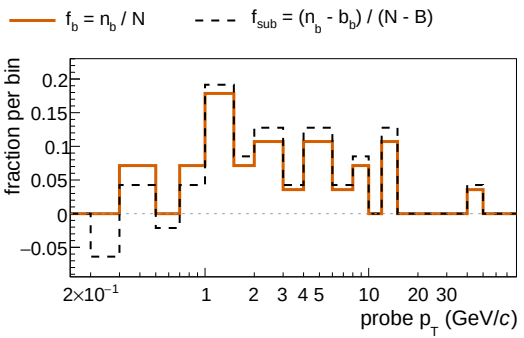
reference: $0.2 \le p_T < 3$ GeV/c

outside jet and perp. cones

ρ = 4.51 GeV/c

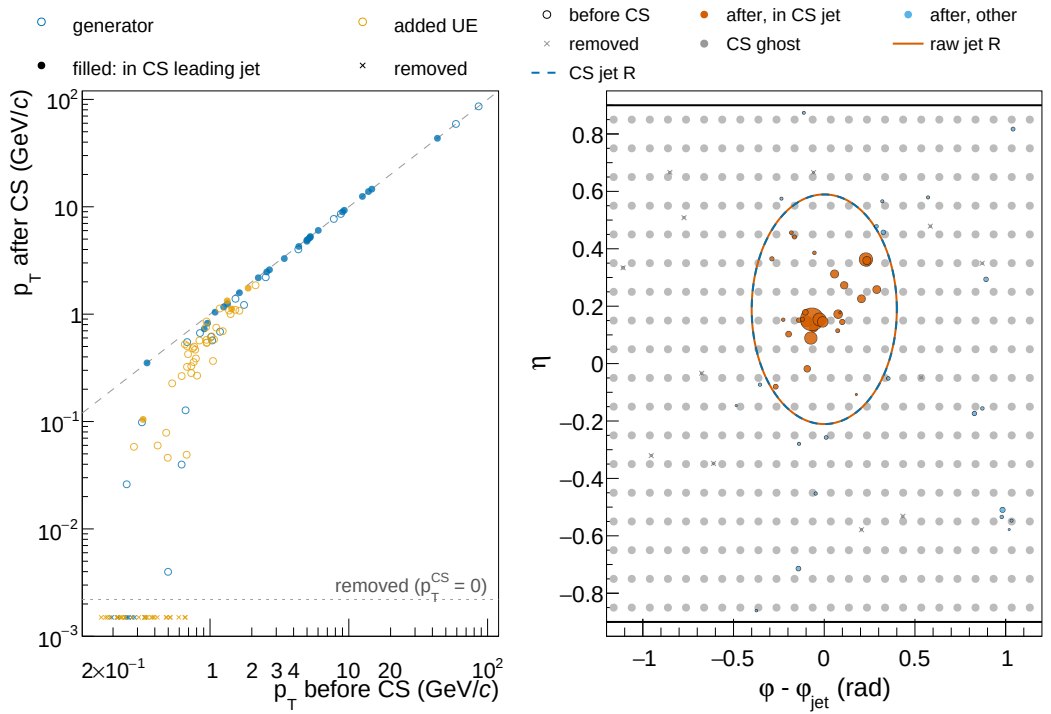
N_{ref} = 71, $\langle p_T \rangle_{ref}$ = 0.732





(c) Jet cone and perpendicular cones ($\varphi_{jet} \pm \pi/2$), track roles; right: in-cone counts n_b, perpendicular-cone estimate b_b, n_b - b_b and the fractions f_b, f_{sub} per probe-p_T bin.

PYTHIA 8 pp $\sqrt{s}$ = 5.36 TeV, event 4, generator particles + thermal UE $dN/d\eta$ = 42, $|\eta_{trk}| < 0.9$, anti-k_T $R = 0.4$



leading jet, R = 0.4

raw p_T = 160.85 GeV/c

area A = 0.555, ρ = 4.51 GeV/c

ρA-corrected p_T = 158.35 GeV/c

CS jet p_T = 158.53 GeV/c

truth (no added UE) p_T = 156.02 GeV/c

tracks in the acceptance

Σ p_T before CS = 385.9 GeV/c

Σ p_T after CS = 353.4 GeV/c

removed: 36 of 118 tracks (11.8 GeV/c)

subtracted Σ p_T = 32.5 GeV/c

(ρ_{CS} × acceptance = 51.0 GeV/c;
ghosts farther than ΔR_{max} keep their p_T)

(d) Event-wise constituent subtraction (ΔR_{max} = 0.25, α = 0, ghost area 0.01, ρ_{CS} = 4.51 GeV/c): track p_T before/after, CS ghost grid near the jet, jet p_T definitions.